

Chapter 5

Basic Computer and Information Theory

Every act of measurement produces data, and data must be encoded before it can be stored, transmitted, or analyzed. The number system used for encoding determines what can be represented exactly and what must be approximated. This chapter develops the positional structure of the base-ten number system, extends it to the binary system used by computers, and connects both to ideas from earlier chapters. Sampling, computing a mean, and compressing a file are all acts of encoding: each discards some information to preserve what the observer needs. The chapter closes with an introduction to information content as a measurable quantity linked directly to probability.

5.1 Base Ten and Positional Notation

The number system in everyday use is the **base-ten** system, also called the **decimal system**. In this system, the value of each digit depends on its position in the number. Each position represents a power of ten.

Definition. *Positional notation.* In **positional notation**, the value of a digit is the digit multiplied by the place value of its position. In base ten, the place values from right to left are $10^0, 10^1, 10^2, 10^3, \dots$, that is, 1, 10, 100, 1000, $\dots$

Example 5.1.1. Expand 4,726 using positional notation.

$$4,726 = 4 \times 1000 + 7 \times 100 + 2 \times 10 + 6 \times 1.$$

4	7	2	6
10^3 = 1000	10^2 = 100	10^1 = 10	10^0 = 1

The place values are a multiplicative recursion with base 10, reading from right to left: 1, 10, 100, 1000, $\dots$. This is the multiplicative recursion from Chapter 1, embedded in the structure of the number system itself.

Decimal Fractions

The decimal point extends positional notation to the right. Each position to the right of the decimal point represents a negative power of ten.

Position	Name	Power	Value
First right	Tenths	10^{-1}	0.1
Second right	Hundredths	10^{-2}	0.01
Third right	Thousandths	10^{-3}	0.001

Example 5.1.2. Expand 0.374 using positional notation.

$$0.374 = 3 \times \frac{1}{10} + 7 \times \frac{1}{100} + 4 \times \frac{1}{1000} = \frac{3}{10} + \frac{7}{100} + \frac{4}{1000}.$$

As a single fraction:

$$0.374 = \frac{374}{1000} = \frac{187}{500}.$$

Example 5.1.3. Every terminating decimal is a rational number. $0.6 = \frac{6}{10} = \frac{3}{5}$.

A decimal that repeats without terminating is also rational. $0.333\ldots = \frac{1}{3}$.

A decimal that neither terminates nor repeats is irrational. $\pi = 3.14159\ldots$ has no fraction representation.

Exercises 5.1

5.1.1. Expand each number using positional notation.

- (a) 3,509
- (b) 81,042
- (c) 0.625
- (d) 0.083

5.1.2. Write each decimal as a fraction in simplest form.

- (a) 0.4
- (b) 0.75
- (c) 0.125
- (d) 0.36

5.1.3. Classify each decimal as rational or irrational. Justify each answer in one sentence.

- (a) $0.5000\dots$ (terminates)
- (b) $0.142857142857\dots$ (repeats)
- (c) $1.41421356\dots$ ($=\sqrt{2}$, never repeats)
- (d) $0.250000\dots$ (terminates)

5.1.4. The number $\frac{1}{7} = 0.142857142857\dots$ repeats with a period of 6 digits. Write the first 12 digits of this decimal. Then verify by computing $\frac{1}{7}$ on a calculator and comparing.

5.2 Binary Numbers

A computer stores all information using exactly two symbols: 0 and 1. A single such symbol is called a **bit**. The number system built from bits is **base two**, called the **binary system**.

Definition. *Binary number.* A **binary number** uses positional notation with base 2. Each position represents a power of 2. The place values from right to left are $2^0, 2^1, 2^2, 2^3, \dots$, that is, 1, 2, 4, 8, 16, 32, $\dots$

The place values of the binary system are the doubling recursion from Chapter 1: 1, 2, 4, 8, 16, 32, $\dots$. This recursion now serves as the structural basis of computer arithmetic.

Converting Binary to Base Ten

To convert a binary number to base ten, multiply each bit by its place value and add.

Example 5.2.1. Convert 1011_2 to base ten.

1	0	1	1
2^3	2^2	2^1	2^0
$= 8$	$= 4$	$= 2$	$= 1$

$$1011_2 = 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 8 + 0 + 2 + 1 = 11_{10}.$$

Example 5.2.2. Convert 10110_2 to base ten.

$$\begin{aligned} 10110_2 &= 1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1 \\ &= 16 + 0 + 4 + 2 + 0 = 22_{10}. \end{aligned}$$

Converting Base Ten to Binary

To convert a base-ten integer to binary, divide repeatedly by 2 and record the remainders. Reading the remainders from bottom to top gives the binary representation.

Example 5.2.3. Convert 13_{10} to binary.

Division	Quotient	Remainder
$13 \div 2$	6	1
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

Reading remainders from bottom to top: 1101_2 .

Check: $1 \times 8 + 1 \times 4 + 0 \times 2 + 1 \times 1 = 13$. ✓

Example 5.2.4. Convert 25_{10} to binary.

Division	Quotient	Remainder
$25 \div 2$	12	1
$12 \div 2$	6	0
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

Reading from bottom to top: 11001_2 .

Check: $16 + 8 + 0 + 0 + 1 = 25$. ✓

Exercises 5.2

5.2.1. Convert each binary number to base ten.

(a) 101_2

(b) 1110_2

(c) 10001_2

(d) 11111_2

5.2.2. Convert each base-ten number to binary using repeated division.

(a) 7

(b) 18

(c) 31

(d) 40

5.2.3. The binary number 11111111_2 has eight bits (one byte).

(a) Convert this to base ten.

(b) What is the largest base-ten number that can be stored in one byte? (*Hint: a byte holds 8 bits, each either 0 or 1.*)

5.2.4. A byte stores the binary number 01000001_2 .

(a) Convert to base ten.

(b) In the ASCII encoding standard, this value corresponds to the letter A. Without looking it up, what base-ten value would you expect for the letter B? Convert your prediction to binary.

5.3 Binary Fractions and Approximation

Just as base ten extends to the right of the decimal point, binary extends to the right of the **binary point**. Positions to the right represent negative powers of 2.

Binary	Meaning	Fraction	Decimal
0.1_2	$\frac{1}{2}$	$\frac{1}{2}$	0.5
0.01_2	$\frac{1}{4}$	$\frac{1}{4}$	0.25
0.001_2	$\frac{1}{8}$	$\frac{1}{8}$	0.125
0.0001_2	$\frac{1}{16}$	$\frac{1}{16}$	0.0625

Example 5.3.1. Convert 0.101_2 to a base-ten decimal.

$$0.101_2 = 1 \times \frac{1}{2} + 0 \times \frac{1}{4} + 1 \times \frac{1}{8} = 0.5 + 0 + 0.125 = 0.625.$$

When Binary Cannot Be Exact

Many base-ten fractions have no exact binary representation. The fraction $\frac{1}{10} = 0.1$ in base ten requires an infinite repeating binary string:

$$0.1_{10} = 0.0001100110011 \dots_2.$$

Every computer that stores 0.1 is storing an approximation. This is why arithmetic software sometimes returns results such as:

$$0.1 + 0.2 = 0.30000000000000004.$$

The computation is correct. The base-two encoding of $\frac{1}{10}$ cannot be exact, and the error accumulates during addition.

Fraction	Exact in binary?	Reason
$\frac{1}{2}$	Yes	0.1_2
$\frac{1}{4}$	Yes	0.01_2
$\frac{3}{8}$	Yes	0.011_2
$\frac{1}{5}$	No	$0.001100110011 \dots_2$
$\frac{1}{10}$	No	$0.0001100110011 \dots_2$
$\frac{1}{3}$	No	$0.010101 \dots_2$

A fraction is exactly representable in binary if and only if its denominator (in simplest form) has no prime factors other than 2.

Example 5.3.2. Is $\frac{3}{16}$ exactly representable in binary? $16 = 2^4$. The denominator's only prime factor is 2. Yes: $\frac{3}{16} = 0.0011_2$.

Check: $0 + 0 + \frac{1}{8} + \frac{1}{16} = \frac{2}{16} + \frac{1}{16} = \frac{3}{16}$. ✓

Is $\frac{1}{6}$ exactly representable in binary? $6 = 2 \times 3$. The denominator has 3 as a prime factor. No exact binary representation exists.

Exercises 5.3

5.3.1. Convert each binary fraction to a base-ten decimal.

- (a) 0.1_2
- (b) 0.11_2
- (c) 0.101_2
- (d) 0.1001_2

5.3.2. Determine whether each fraction has an exact binary representation. Justify your answer using prime factorization of the denominator.

(a) $\frac{7}{8}$

(b) $\frac{3}{10}$

(c) $\frac{5}{32}$

(d) $\frac{1}{12}$

5.3.3. A student types $0.1 + 0.2$ into a calculator and gets 0.30000000000000004. Explain in two complete sentences why this happens, using the ideas from this section.

5.3.4. Financial software often stores dollar amounts as integers (cents) rather than decimals to avoid binary approximation errors. Explain in one sentence why storing \$4.37 as 437 cents eliminates the problem described in this section.

5.4 Data as Encoding

Every measurement is a choice about what information to retain. The observer selects a level of precision, a set of categories, or a summary statistic, and in doing so discards everything not captured by that choice. This is encoding: the transformation of raw observations into a structured representation.

This process has appeared in every chapter of this course.

Chapter	Raw data	Encoding
2	List of responses	Frequency table
2	Frequency table	Probability distribution
3	Data set	Mean and standard deviation
4	Population	Sample

Each step in this table reduces information. The frequency table discards the order of responses. The mean discards the shape of the distribution. The sample discards observations not selected. This is not a flaw. It is the purpose of encoding: to retain what is needed and discard what is not.

Compression

Definition. *Data compression.* **Data compression** reduces the number of bits needed to store or transmit a data set by removing redundancy.

Run-length encoding is a simple compression method. Instead of storing each symbol individually, it stores each distinct symbol together with the number of times it appears consecutively.

Example 5.4.1. The string AAAAAABBBCC has 11 symbols. Using run-length encoding: 6A3B2C. This representation uses 6 characters instead of 11.

The string ABCABCABC has 9 symbols. Using run-length encoding: 1A1B1C1A1B1C1A1B1C. This is longer than the original. Run-length encoding is efficient for data with long runs of repeated values and inefficient for highly varied data.

The efficiency of compression depends on the structure of the data. A data set with low variability (small standard deviation) contains more repetition and compresses well. A data set with high variability (large standard deviation) contains less repetition and compresses poorly. This connects compression directly to the spread measures of Chapter 3.

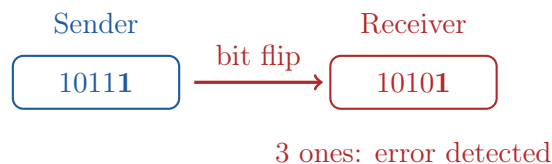
Error Correction

When data is transmitted over a noisy channel, individual bits may be corrupted: a 0 may become a 1 or vice versa. **Error correction** adds strategic redundancy to the data so that the receiver can detect and repair corruption.

Definition. *Parity bit.* A **parity bit** is an extra bit appended to a block of data. Its value is chosen so that the total number of 1s in the block (including the parity bit) is even. If the receiver counts an odd number of 1s, at least one bit was corrupted in transit.

Example 5.4.2. A sender transmits the block 1011. This block contains three 1s (odd). The parity bit is set to 1 so that the total count of 1s becomes four (even). The transmitted block is 10111.

During transmission, one bit is corrupted: the receiver gets 10101. Counting 1s: there are three (odd). The receiver detects an error.



Note that the parity bit detects the error but cannot identify which bit was corrupted. More sophisticated error-correcting codes can both detect and repair errors.

Exercises 5.4

5.4.1. Apply run-length encoding to each string. State whether the encoding is shorter than the original.

- (a) HHHHHTTTTT
- (b) XYZXYZXYZ
- (c) AAABBAAA
- (d) MMMMMMMMM

5.4.2. A data set of test scores has standard deviation $s = 1.2$. Another has standard deviation $s = 18.4$.

- (a) Which data set is more likely to compress well? Explain using the concept of variability.
- (b) In one sentence, state the general relationship between standard deviation and compression efficiency.

5.4.3. For each block, determine the correct parity bit and write the full transmitted block.

- (a) 1100
- (b) 1010
- (c) 0111

5.4.4. A receiver gets the following blocks. For each, determine whether an error is detected.

- (a) 10110 (parity bit is the last bit)
- (b) 11011
- (c) 00000

5.5 Information Content

In everyday usage, the word “information” refers broadly to knowledge or data. In information theory, it has a precise mathematical meaning connected directly to probability.

Definition. *Information content.* The **information content** of an outcome with

probability p is:

$$I = \log_2 \left(\frac{1}{p} \right) \text{ bits.}$$

An outcome with probability $\frac{1}{2}$ carries 1 bit of information. An outcome with probability $\frac{1}{4}$ carries 2 bits. An outcome with probability $\frac{1}{2^n}$ carries n bits.

Since this course does not assume familiarity with logarithms, the table below gives the information content for probabilities that are powers of $\frac{1}{2}$. The pattern is: each time the probability halves, the information content increases by exactly 1 bit.

Probability	Information (bits)	Intuition
1	0	Certain outcome carries no information
1/2	1	Fair coin flip
1/4	2	One of four equally likely outcomes
1/8	3	One of eight equally likely outcomes
1/16	4	One of sixteen equally likely outcomes
1/32	5	One of thirty-two equally likely outcomes
1/1024	10	One of 1,024 equally likely outcomes

The connection to binary numbers is exact: an outcome that identifies one of 2^n equally likely possibilities carries n bits, because n binary digits are needed to encode which outcome occurred.

Example 5.5.1. A fair coin has two equally likely outcomes: heads and tails. Each outcome has probability $\frac{1}{2}$ and carries 1 bit of information.

A fair four-sided die has four equally likely outcomes. Each outcome has probability $\frac{1}{4}$ and carries 2 bits of information.

A standard eight-sided die has eight equally likely outcomes. Each outcome has probability $\frac{1}{8}$ and carries 3 bits of information.

Example 5.5.2. A weather forecast predicts rain tomorrow with probability $\frac{1}{2}$. If it rains, the outcome carries 1 bit of information.

If instead the forecast had assigned probability $\frac{1}{8}$ to rain, and it rained, the outcome would carry 3 bits of information. The rarer the event, the more information is conveyed when it occurs.

Example 5.5.3. (*Information and probability distributions*) Consider two probability distributions over four outcomes.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
Distribution 1 (uniform)	1/4	1/4	1/4	1/4
Distribution 2 (concentrated)	1/2	1/4	1/8	1/8

Under Distribution 1, every outcome carries 2 bits. Under Distribution 2, outcome *A* carries 1 bit (it is common, so its occurrence is less surprising) while outcome *D* carries 3 bits (it is rare, so its occurrence conveys more information).

The total information in a distribution is the average information per outcome, weighted by probability. Distribution 1 has higher average information because its outcomes are more evenly distributed and therefore more uncertain.

Exercises 5.5

5.5.1. Use the information table to find the information content of each outcome.

- (a) $P(\text{outcome}) = 1/4$
- (b) $P(\text{outcome}) = 1/32$
- (c) $P(\text{outcome}) = 1$
- (d) $P(\text{outcome}) = 1/16$

5.5.2. A bag contains 8 colored chips, all different colors. One chip is drawn at random.

- (a) What is the probability of drawing any particular chip?
- (b) How many bits of information does drawing any particular chip carry?
- (c) How many binary digits would be needed to encode which chip was drawn?

5.5.3. A data set has outcomes with probabilities $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, and $\frac{1}{8}$.

- (a) Verify this is a valid probability distribution.
- (b) Find the information content of each outcome.
- (c) Which outcome carries the most information? The least?

5.5.4. In one complete sentence, explain why a certain outcome (probability = 1) carries zero bits of information. Connect your answer to the role of probability in measuring surprise.

Chapter 5 Review

5. Expand 7,304 using positional notation. Then write 0.48 as a fraction in simplest form.
5. Convert each number as indicated.
 - (a) 1101_2 to base ten
 - (b) 10101_2 to base ten
 - (c) 19_{10} to binary
 - (d) 44_{10} to binary
5. Convert 0.1101_2 to a base-ten decimal. Then determine whether $\frac{3}{20}$ has an exact binary representation. Justify using prime factorization.
5. The following data records the daily high temperature (in degrees F) for 10 days:

$72, 72, 72, 73, 71, 72, 72, 85, 72, 72.$

 - (a) Compute the mean and standard deviation.
 - (b) Apply run-length encoding to the string: TTTTTTTTFF (T = temp within 1 degree of 72, F = outside that range).
 - (c) Is this data set likely to compress well? Relate your answer to the standard deviation.
5. A transmitted block is 01101. The last bit is the parity bit.
 - (a) Count the number of 1s in the block.
 - (b) Was an error detected? Explain.
5. A lottery has 32 equally likely outcomes.
 - (a) What is the probability of any one outcome?
 - (b) How many bits of information does winning carry?
 - (c) How many binary digits are needed to encode which outcome occurred?
5. A probability distribution over three outcomes has probabilities $\frac{1}{2}$, $\frac{1}{4}$, and p . Find p . Then find the information content of each outcome.
5. Explain in three complete sentences how the concept of encoding connects the topics of this chapter (base ten, binary, compression, error correction, and information content) to the statistical tools developed in Chapters 2 through 4.